

Car Crashes

(in Colorado)

Eytan Abramowitz

The Idea

Prevent Car Accidents using Data Science

- ~38,000 fatalities & 6 million injuries from car accidents per year (USA)
- Understanding causes/correlations can help drivers avoid dangerous situations

The Data

Car Crash Data

- 7.7 Million crashes in the US from 2016-2023
- Contains info on location, weather conditions, traffic controls
- Source: kaggle.com

Public School Data

- Basic information on US & territories public schools
- Has location and school name
- Source: National Center for Education Statistics

'Synthetic' Weather Data

- Random weather points
- Explained more later
- Source: open-meteo.com

Initial Cleaning

Car Crash Data

- Only use data from Colorado
- Remove unnecessary columns

Public School Data

- Only use data from Colorado
- Only if the name contains 'High'
- Just keep name, latitude, longitude columns

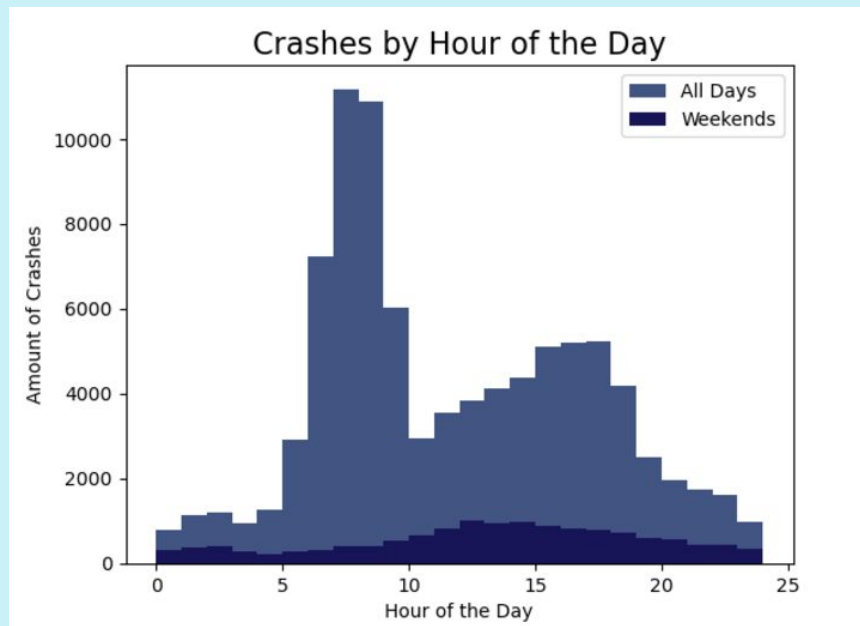
'Synthetic' Weather Data

- N/a

Investigation 1 - Times

Time of Day

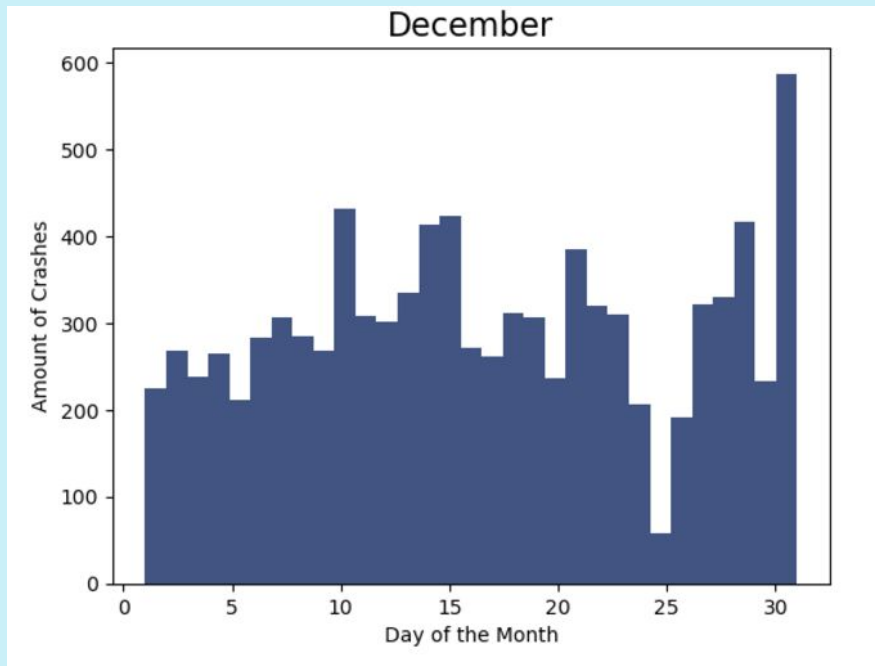
- Just using crash data
- Convert time data to DateTime obj
- Histogram reveals two spikes
- Is this because of twilight conditions?
Rush-hour?
- Trend disappears on weekends



Investigation 1 - Times

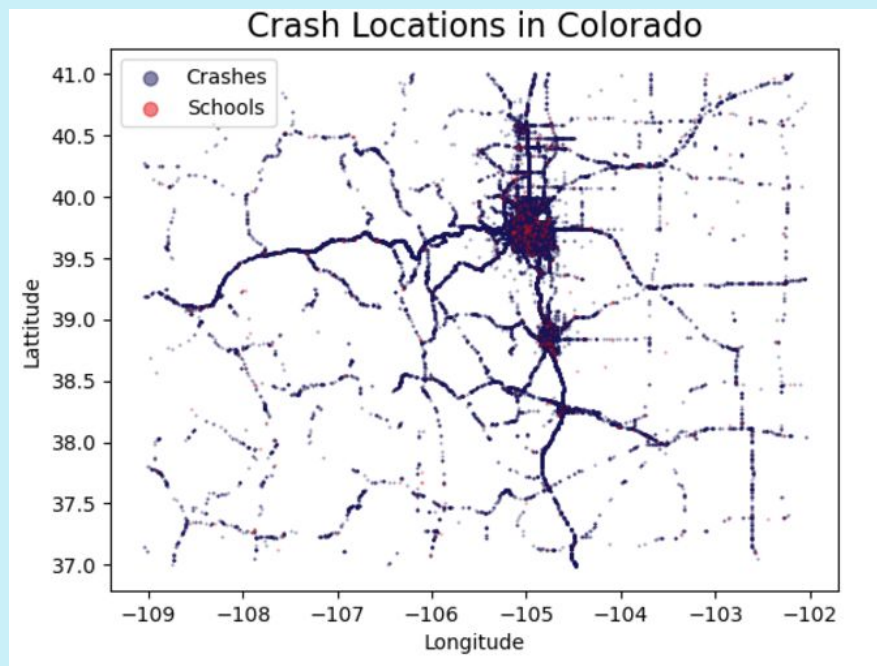
Time of Year

- Use histogram to quickly visualize every day of the year
- December has the days with greatest and fewest crashes (excluding 2/29).
- Christmas has the fewest crashes (59)
- New Year's eve has the most (589)



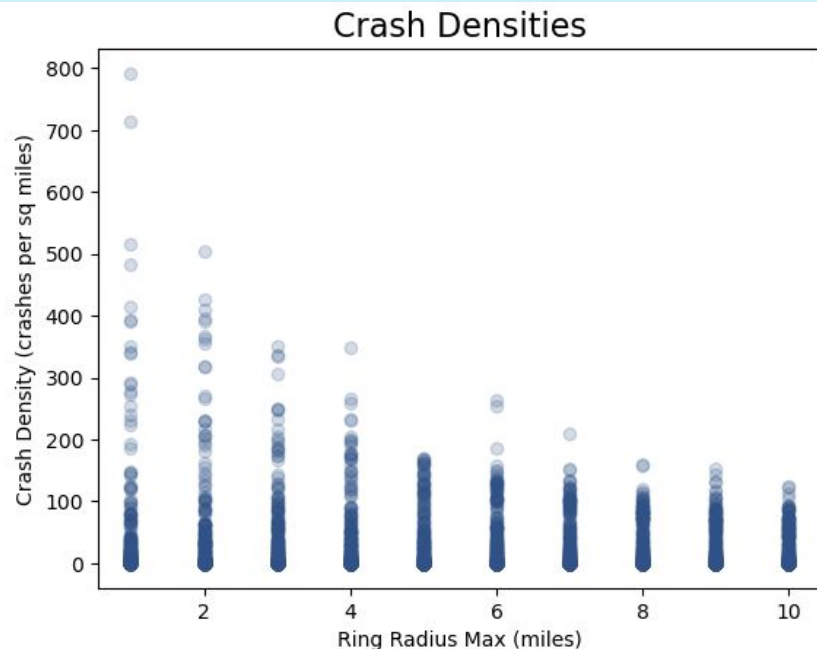
Investigation 2 – Locations

- Are there certain locations that are more dangerous to drive in?
- Initial plot intuitively shows highways and urban centers have more crashes
- Maybe areas near schools which are centers for young drivers are also more dangerous



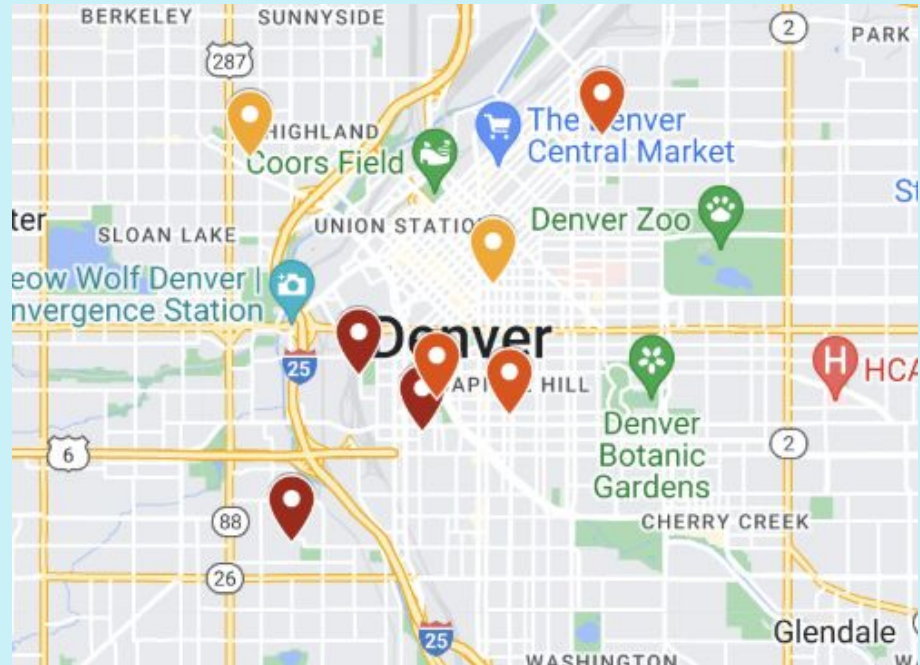
Investigation 2 - Locations

- Reformat the data so that vectorized functions can be used (~350 schools, ~90,000 crashes)
- Use modified distance formula to find distance between each school and each crash
- Create 'rings' to group data like a histogram
- Calculate crash density per ring for every school
- Plot densities - Correlation! (maybe)



Investigation 2 - Locations

- Took schools with the highest densities for radii 1-3 miles.
- They are all close to each other and to major highway intersections
- Need to do more to better understand the correlation.
- Still possible these highway intersections are worse because they are near schools. Cause is unclear.



Investigation 3 - Weather

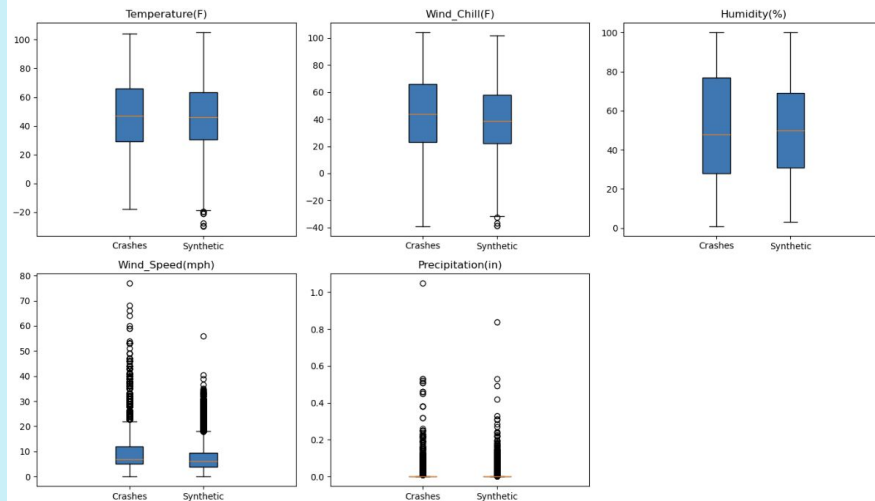
- Intuition says that bad weather leads to more crashes
- Does the data agree?
- If it does how much more dangerous are certain conditions?
- Drop rows with NA weather values

- The crash data only provides 'positives.' I want to compare to 'negative' crash events
- Found an API that allows me compile historical weather data
- Collected ~50,000 random weather points ranging from:
 - LATs: [37, 41]
 - LONs: [-110,-102]
 - Dates: [3/1/2016, 3/31/2023]

Investigation 3 – Weather

Is there a difference between crashes & synthetic?

- Compare the distribution of different weather variables for crashes vs. synthetic
- Use stats library to test t-statistic and p-value for the different groups
- Run logistic regression
 - Get log odds for each input
- Run decision tree



Logistic Regression:

Accuracy = 0.58025520483546

Intercept = [-1.52831564]

Predictors = ['Wind_Chill(F)', 'Humidity(%)', 'Wind_Speed(mph)', 'Precipitation(in)']

Beta Values = [[0.01465126 0.01256021 0.0456685 -6.58926946]]

Odds increase per unit change = [[1.01475911 1.01263942 1.04672736 0.00137504]]

[[11751 11077]

[8923 15897]]

Decision Tree:

Train

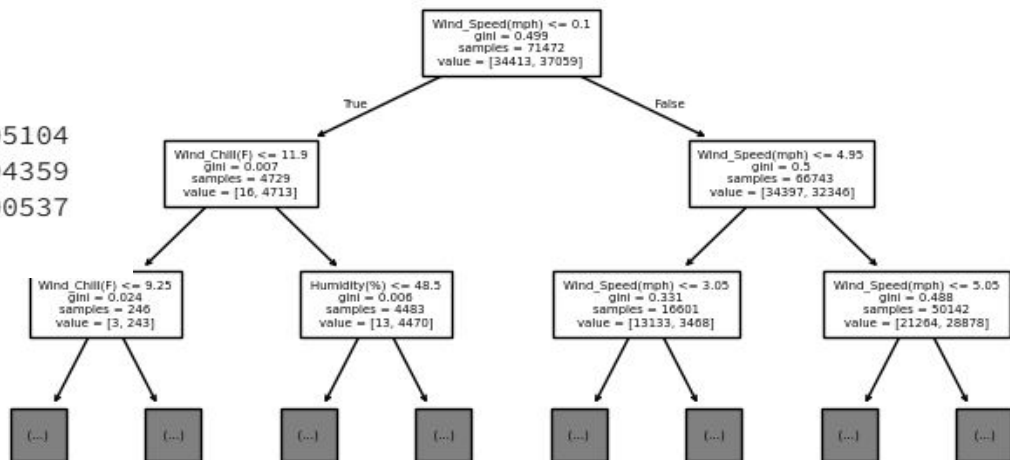
Test

Tree1: 0.9992024848891874, 0.9617612491605104

Tree2: 0.9599283635549586, 0.9583823035594359

Tree3: 0.9553671367808373, 0.9548564472800537

- Seems too good
- Further investigation required



Limitations:

- Lack of car/driving density
- Looked at data from one state. Laws and norms can vary from state to state
- Differing sources of weather data

Overall Conclusion:

- The largest factor by far is (unsurprisingly) where people are
- Avoid traveling if possible when/where other people are
- Need more information about proximity to schools